SAFE-FOOD / OPTION A

A synthetic two-source knowledge graph and unchanged GraphSAGE-style learning experiment | Technical report draft | 19 October 2026

Course: MSc Knowledge Representation Team: add confirmed members and contributions before submission.

Data integrity statement: Both CSV cohorts are synthetic, teaching-only records shaped after Open Food Facts and USDA FoodData Central. No real products or independent external-source matches are claimed. Deliberate injected errors and matching barcode strings are documented. This is not a food-safety or packaging-law decision service.

Executive summary

SAFE-FOOD combines relational data preparation, an application ontology with selected FoodOn identifiers, RDF materialization, SHACL validation, cross-source-shaped identity links, SPARQL access and exploratory graph learning. A separate prepared dataset is created from the raw synthetic CSVs; no valid nutrient measurement or Nutri-Score label is changed merely to improve model performance.

Dataset measure	Raw	Prepared
OFF-shaped products	3,001 rows (3,000 unique codes)	2,925 products
USDA-shaped products	800 rows	797 products
Exact synthetic barcode overlaps	150	146
RDF after preparation	-	56,474 triples; 3,722 typed products
Nutri-Score labels for learning	-	1,945 labelled OFF nodes

1. Problem statement

Research question: Can an ontology-backed data pipeline represent two heterogeneous, synthetic packaged-food record sets, surface source discrepancies and rule-based review flags, validate deterministic constraints and support explainable SPARQL queries, while a pre-existing brand-neighbor GNN classifies reported Nutri-Score groups?

The work tests an integration architecture on constructed examples. It does not establish entity-resolution precision on real branded foods, verified product safety, causal nutritional relationships or generalization of the GNN to real data.

2. Source and preparation policy

Original CSVs remain unchanged under data/raw/. The transparent Python preparation process removes rows with missing required identifiers/names, duplicate OFF identifiers or non-finite/out-of-bounds numerical features. Invalid Nutri-Score Z and invalid NOVA codes are set to missing rather than reassigned. Missing optional numeric values are retained. Cleaning actions, raw checksums and potential same-code matches are stored in data/quality/.

3. Architecture and mapping

The reproducible data path is: raw synthetic CSVs -> prepared synthetic CSVs -> SQLite tables -> RDFLib / R2RML mapping -> SHACL gate -> VoID metadata -> Fuseki SPARQL endpoint -> question/answer access. Machine learning reads the RDF materialization independently.

SQLite retains distinct off_products and usda_products, with off_allergens and off_traces as normalized relational tables. The OWL application ontology in ontology/foodsafe.ttl defines PackagedProduct, Allergen, DataSource, NutriScoreGrade, LabelIncomplete and related properties. ontology/foodon-slim.ttl is a small identifier subset, not an entire FoodOn ontology. Ontology files represent semantics; SHACL files define separate closed-world data constraints.

Mapping/materialize.py creates 56,474 RDF triples from the prepared SQLite data. Each synthetic source keeps its own product IRI. 146 exact barcode pairs become 292 directed foodsafe:sameProductAs statements, one in each direction. Declarative mapping/mapping.ttl now includes both directions. Morph-KGC parity for this new dataset remains to be independently tested.

4. Quality validation and interpretation

The SHACL gate requires a nonempty barcode and name, checks energy in [0,900] kcal/100g and fat in [0,100] g/100g, and validates known grade objects. Independent local RDF sanity checks found no violations of these mirrored required or numerical bounds. The full pySHACL engine could not be executed in the local sandbox because the dependency was unavailable; verification on Windows is required. The raw errors are intentionally retained in the separate original CSVs to demonstrate failing conditions.

Ingredient/allergen comparisons use pattern rules and create hasLabelGapFlag review candidates. A rule flag is not proof that a consumer food label is incorrectly printed, nor a clinical allergen-safety finding. Cross-source nutrition differences are only review candidates, not corrections.

5. Graph learning: original code unchanged

The learning task remains binary node classification: class 1 for reported Nutri-Score A/B, class 0 for C/D/E. USDA synthetic records have no grade labels and are excluded from classification loss but can contribute features through same-brand neighbors. Inputs: energy, fat, sugar, protein, salt. Train-only median imputation and standardization are shared across baseline and GNN; brand-disjoint group splits use seeds 11, 23 and 37.

GraphSAGE-style forward propagation aggregates features over brand-neighbor edges only, and does not consume sameProductAs or Product-Allergen edges for classification. The architecture, task, training epochs and model files were not modified.

Seed	Baseline Macro-F1	GNN Macro-F1
11	0.5062	0.5211
23	0.4830	0.4957
37	0.5293	0.5038
Three-seed mean	0.506	0.507

No substantive GNN advantage is demonstrated. The synthetic Nutri-Score labels were generated without a realistic grading procedure; weak predictability is therefore expected, and these metrics are unsuitable for real-world nutritional claims.

6. Access, deployment and evidence gaps

The Docker Compose stack has a one-shot Python mapping and SHACL pipeline, a Fuseki endpoint for SPARQL and an English-question access layer. Offline answers are deterministic examples, NOT actual Gemini model evaluations. Gemini Live requires a privately configured API key and a fresh evaluation on the new dataset. Project secrets are not included.

Locally executed: data preparation, SQLite generation, Python RDF materialization, required/numeric sanity checks, GNN and baseline training across three seeds, link ranking, VoID generation and non-pySHACL tests. Not yet independently verified on Option A: real pySHACL gate, full Morph-KGC R2RML parity, Docker/Fuseki stack, Gemini Live evaluation, Protégé reasoner consistency and Windows clean run.

7. Limitations and next work

All records and matching identities are synthetic; intentionally planted overlaps do not measure genuine data integration accuracy. The data cleaning policy is recorded and does not restore the true values of discarded invalid observations. Nutri-Score labels have weak substantive realism and do not support generalizable graph-learning claims. Future work requires real OFF and USDA Branded Foods downloads with GTIN/packaging-version checks, human-reviewed nutritional/allergen annotations and independent evaluation.

Reproducibility: RUN_GUIDE_AR.md; prepare script: data/prepare_option_a.py; quality audit: data/quality/summary.json; ontology: ontology/foodsafe.ttl; mapping: mapping/mapping.ttl; training source: model/gnn.py (unchanged).